# INTRODUCTION

The previous learning module described issuers' *short-term* investments and financing activities. In this module and the next, we turn our attention to issuers' *long-term* investment and financing activities. First, we explore the various forms of capital investment and their purposes. We then discuss the investment decision-making process and compare analytical approaches employed in that process. In the third lesson, we describe principles of capital allocation and common pitfalls. While the goals of both capital allocation and estimating expected investment returns are to select the best choice among investment alternatives, a firm's decision today may influence future investment decisions, resulting in so-called real options, which are discussed in the final lesson.



## Learning Module Overview

- Companies make capital investments to maintain or expand operations. Capital investments can be grouped into four categories based on their risk and return characteristics: (1) going concern projects, (2) regulatory/compliance projects, (3) expansion projects, and (4) other.
- Capital allocation is a process undertaken by issuers' management and board for evaluating investment opportunities based on their expected contribution to shareholder value, as well as other considerations, such as environmental, social, and governance (ESG) factors. Although some projects might look profitable on an accounting or standalone basis, they might be uneconomical compared to alternatives or from an overall strategic perspective. Such projects should not be pursued, and capital should instead be returned to shareholders.
- Net present value (NPV) and internal rate of return (IRR) are two tools used to evaluate individual investment projects. NPV estimates the increase in firm value from a project, while IRR is an estimate of the rate of return on a project, subject to certain assumptions, which can be compared to a hurdle rate.
- Unlike NPV and IRR, return on invested capital (ROIC) is a company-wide measure and can be calculated using data available to independent analysts. ROIC is the rate of return an issuer earns over a period across all investments and can be compared to an investor's required rate of return. Like NPV and IRR, ROIC is subject to limitations and assumptions.
- Before investment projects are appraised on a quantitative basis, they should be modeled in accordance with certain principles, including measurement of cash flows on an after-tax basis, avoiding double counting, and including a project's impact on the rest of the firm. Impacts can be positive, such as cost savings, or negative, such as the loss of sales from existing products.
- Apart from deviations from these principles, capital allocation is additionally prone to behavioral biases and cognitive errors. These pitfalls can be detected by a thorough analysis of a company's financials on a historical and comparative basis, as well as an examination of corporate governance and management renumeration policies.
- Real options are like financial options in that they provide a right, not an obligation, for management to alter different aspects of capital projects in the future. Those aspects include timing and size of a project, as well as flexibility with regard to future pricing policies or operating capacity.

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More advanced methods include decision trees and option pricing models, which require assumptions about the probability of future events.

## Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

**1.** The following list contains either an example or an attribute of a capital investment project. Assign each item to either *maintenance* or *growth*.

- Acquisition
- Expand business scope
- Research and development
- Replace outdated facilities
- High-risk investment and uncertainty
- Limited downside risk and uncertainty
- Needed to meet safety, compliance, regulatory standards

| Maintenance | Growth |
|---|---|
| | |

Hide Solution

**Solution:**

| Maintenance | Growth |
|---|---|
| Replace outdated facilities | Acquisition |
| Needed to meet safety, compliance, regulatory standards | Expand business scope |
| Limited downside risk and uncertainty | Research and development |
| | High-risk investment and uncertainty |

**2.** When calculating IRR, the interim cash flows are assumed to be reinvested and earn a rate of return rate that is:

**A.** lower than IRR.
**B.** the same as IRR.
**C.** higher than IRR.

Hide Solution

**Solution:**

B is correct. An important assumption of IRR is that it represents only the (geometric) rate of return on the investment if interim cash flows are reinvested at the IRR.

A is incorrect because if reinvestment rates are lower compared to IRR, the rate of return on the investment will be lower than the IRR.

**3.** Complete the following sentences by filling in the blanks using the terms provided.

When calculating ROIC, an independent analyst should add _____ and _____ to calculating average invested capital. ROIC, _____ project NPV and IRR, can be calculated using data available to independent investment analysts.

like

unlike

equity

short-term assets

long-term assets

long-term liabilities

Hide Solution

**Solution:**

When calculating ROIC, an independent analyst should add *equity* and *long-term liabilities* to calculate average invested capital. ROIC, *unlike* project NPV and IRR, can be calculated using data available to independent investment analysts.

**4.** Explain why capital allocation decisions should *not* be based on accounting measures such as earnings per share (EPS).

Hide Solution

**Solution:**

Capital investments with a positive NPV can reduce rather than increase accounting measures in the near term, while cost cutting and share buybacks, in contrast, may have a positive effect on such measures. Basing investment decisions on short-term accounting numbers can lead a company to choose investments that are not in the long-run interests of its shareholders. Additionally, capital allocation should consider opportunity costs, such as by using a required rate of return for calculating NPV and an appropriate hurdle rate for an IRR. Accounting profits do not consider opportunity costs.

**5.** The annual report of company XYZ contains the following disclosures:

Disclosure 1: "XYZ's management compensation is based on exceeding a target EPS growth rate."

Disclosure 2: "XYZ's management does not change the required rate of return when evaluating capital projects based on whether they are financed by internal or external sources."

Disclosure 3: "When evaluating investment projects, XYZ prepares cash-flow projections based on inflation-adjusted cash flows and discounts them using real rates."

Which of the disclosed policies does *not* conform to best practices regarding capital allocation?

**A.** Disclosure 1
**B.** Disclosure 2
**C.** Disclosure 3

Hide Solution

**Solution:**

Management compensation should incorporate a longer-term perspective and a measure that better considers required rates of return, such as ROIC.

B is incorrect because internally generated capital, such as cash flow from operations, is equity financing and it could be returned to equity investors as a dividend. Regardless of the financing source, management should use appropriate risk-adjusted required rates of return to evaluate capital investments.

C is incorrect because companies may perform analysis in either nominal or real terms, but the approach to cash flows and the discount rate should be consistent. That is, nominal cash flows should be discounted at a nominal discount rate, and real (inflation-adjusted) cash flows should be discounted at a real rate.

**6.** Explain what real options are and how they influence company value.

Hide Solution

**Solution:**

Real options are similar to financial options, except that they deal with real, instead of financial, assets. Real options grant companies the right to make a decision (but do not impose an obligation) in the future that alters the value of capital investment decisions made today. Real options, by providing future decision-making flexibility to companies, can be an important piece of the value in many capital investments.

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